L2 – Modelling a Spring (Hooke's Law)

Unit 1: Oscillations & Waves • MYP Sciences Year 9

Physics 9A and 9B • Mr. Rawson

9A	Mon 31 Aug, P1, 08:55–09:50	**Next**	Thu 3 Sep, P6 — **3 days**
9B	Tue 1 Sep, P2, 09:55–10:45	**Next**	Fri 4 Sep, P3 — **3 days**
Criteria	B (variables, fair test, method) C (table, graph, conclusion) — *formative*	**Lesson**	2 of the taught sequence

To print **26** handouts (13 per set, 3 pages each). One document serves both classes.

Kit Per pair: clamp stand, boss and clamp, spring, mass hanger + 100 g slotted masses, metre rule or 30 cm rule, eye protection. 13 students = 6 pairs and one three, so **7 stations**. [[ALEX: decide — which springs are actually in the cupboard and their constants; the whole design turns on the sets having visibly different stiffnesses]] [[ALEX: decide — confirm the stands, masses and eye protection with Joel before Monday P1]]

Devices The spreadsheet step needs one device per pair. [[ALEX: decide — what the students actually have in PH1 (Chromebooks? iPads? a booking?), and whether the spreadsheet is Google Sheets or Excel; the handout is written tool-neutral so either works]]

Homework None built — the brief was plan + handout only. [[ALEX: decide — whether to set any homework from this lesson]]

Timing The inherited StL unit planner says **40**-minute lessons; the bell schedule and the Class Log clock spans say **50**. Built to 50. Note 9A's P1 is nominally **55** (08:55–09:50) — the spare five belongs in the practical, not in more content.

By the end of the lesson students can:

- measure the **extension** of a loaded spring — not its total length — and record readings in a properly headed table;
- plot force against extension, on paper and in a spreadsheet, and put a **line of best fit** through the points;
- state the relationship in their own words: the graph is a straight line through the origin, so extension is proportional to force;
- name the independent, dependent and control variables of the investigation (Criterion B language).

Timing — run for 9A at P1; 9B deltas below

Time	Phase	What happens
0–5	Do Now	Retrieval from 27 Aug, not new teaching: a hanging mass on the board — "draw the model." Look for the free-body arrows, the fixed point, and gravity as a vector that does not change. This is the hook the whole lesson hangs on: *we already know how to draw a model; today we learn how to build one from data.*
5–12	Launch + demo	Hold up two springs of different stiffness. The question, verbatim: *"Can we predict how far a spring stretches before we pull it?"* Demo one station: clamp, spring, hanger, rule. Define **extension = stretched length** − **natural length** and measure the natural length out loud. Give the force conversion as a tool, not a derivation: each 100 g mass pulls down with about 1 N. Safety: eye protection on, feet clear of the drop zone, and **stop loading** the moment a spring's coils visibly separate at rest — a wrecked spring is a wrecked data set.
12–15	Plan (Criterion B)	Handout Part 2, done *before* touching the kit: independent, dependent and control variables, and the fair-test question. Two minutes, pairs, then take answers cold. Do not move on until "we measure extension, not length" has been said by a student.
15–30	Practical	Pairs collect at least six readings, adding one mass at a time, recording total length and computing extension as they go — not at the end. Circulate hunting for the one error that ruins everything: **recording total length in the extension column**. Fast pairs swap springs and start table two.
30–40	Spreadsheet	Handout Part 5: type the force and extension columns in, insert a scatter chart, add the line of best fit. The chart is the payoff — a straight line the machine agrees with. Pairs that finish plot the paper graph too (Part 4); slower pairs prioritise the spreadsheet and sketch the paper graph from it.
40–47	Conclusion (Criterion C)	Handout Part 6. The sentence to fish for: *"when the force doubles, the extension doubles."* Then the cross-group compare: every pair's graph is a straight line through the origin, but the **slopes differ** — steeper means stretchier. The spring's own personality is one number, the gradient.
47–50	Debrief + name it	Only now: Robert Hooke did exactly this in 1660, and what they have each built is **Hooke's Law**. Close on the transfer question, verbatim, because it is L3's opening line: *"That was a system sitting still. If it is a pendulum going back and forth — how do we model **that**?"*

9B deltas (Tue P2). A true 50 rather than 9A's 55 — hold the launch to seven minutes and protect the spreadsheet block. Adjust from how 9A's practical actually ran on Monday; the Class Log entry is where that note lives.

Questions to ask

1. Why do we write down the natural length before adding any masses? *(Extension is a change — you need the starting point.)*
2. Your first reading is 100 g. Why does the force column say 1 N and not 100? *(The spring feels the pull of gravity on the mass — force in newtons, not mass in grams.)*
3. Two groups tested different springs and both got straight lines — so what is actually different between the springs? *(The gradient. Same law, different number.)*
4. Your line of best fit misses some points. Is that a mistake? *(No — real measurements scatter; the line is the model, the points are the data.)*
5. Can your model predict the extension for a force you never tested? Where would you stop trusting it? *(Interpolation is safe; extrapolation fails when the spring overstretches — every model has a range where it works.)*

Common misconceptions

- **Total length recorded as extension.** The classic. The table forces the subtraction with separate columns, but circulate for it anyway — one wrong column poisons the graph.
- **Mass and force conflated.** Grams go in the mass column, newtons in the force column; the graph is plotted from newtons.
- **Dot-to-dot instead of a line of best fit.** They will join the points. Say it plainly: the ruler line *is* the model.
- **"Is this right?"** Some will want the answer handed over. The answer is their data. Deflect to the graph: "what does your line say?"
- **Overloading.** A spring past its limit gives a curve at the top end. If a group produces one, do not treat it as failure — it is the best teaching moment in the room: the model has a boundary.

Differentiation

Support: the table headings and units are pre-printed, so the recording scaffold is already on the sheet; for the conclusion, offer the sentence frame *"When the force doubles, the extension …"* rather than a blank page.
Extend: second spring, second table — then put both lines on one set of axes and say which spring is stiffer *and how the graph knows*. Sharpest students: predict the extension at a force they did not test, then measure it and judge their own model.

After the lesson

- Write **which springs were used** (and, if measured, their gradients) into this plan and the unit map — L3 and the Criterion B/C pendulum investigation inherit the kit decisions made here.
- Note **how far the spreadsheet step actually got**. If it consumed the lesson, the pendulum lesson's data handling needs a paper-first fallback.
- Note which pairs reached the second spring — that is the extension cohort for the investigation proper.
- 9A runs first: adjust the 9B timings from what P1 actually did.
- Flip both Class Log entries to **Taught**.